

HDOC DAY-7 ACTIVITY

C Programming Report

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Batch	15
Activity	HDOC Day-7
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This report contains the required problem analysis, test cases, algorithms, program evidence, compilation/execution evidence, and evaluation for all three assigned questions.

Question 1 - Mass from Density and Volume

1. Problem Understanding

Input: Density and Volume

Output: Mass

Formula: $\text{Mass} = \text{Density} \times \text{Volume}$

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Density=2, Volume=1	Mass = 2.00	Mass = 2.00	Pass
2	Density=2, Volume=3	Mass = 6.00	Mass = 6.00	Pass
3	Density=7, Volume=89	Mass = 623.00	Mass = 623.00	Pass

3. Algorithm

1. Start.
2. Read density and volume.
3. Calculate $\text{mass} = \text{density} \times \text{volume}$.
4. Display mass.
5. Stop.

Algorithm Evaluation

For all three prepared test cases, applying the formula gives 2.00, 6.00, and 623.00 respectively. These match the expected results, so the algorithm passes all test cases.

4. C Program

The program is attached as a screenshot only, as requested; no plain-text source code is included.

```
UW PICO 5.09 File: 7.1.c Modified
#include <stdio.h>

int main()
{
    float density, volume, mass;

    printf("Enter density: ");
    scanf("%f", &density);

    printf("Enter volume: ");
    scanf("%f", &volume);

    mass = density * volume;

    printf("Mass = %.2f\n", mass);

    return 0;
}
```

Program screenshot for Question 1 (file 7.1.c).

5-6. Compilation, Execution and Code Evaluation

```
Last login: Mon Aug 31 23:12:06 on tty900
dhrupandey@DHRUVs-MacBook-Air ~ % nano 7.1.c

dhrupandey@DHRUVs-MacBook-Air ~ % clang 7.1.c -o7.1
dhrupandey@DHRUVs-MacBook-Air ~ % ./7.1.c
zsh: permission denied: ./7.1.c
dhrupandey@DHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 1
Mass = 2.00
dhrupandey@DHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 3
Mass = 6.00
dhrupandey@DHRUVs-MacBook-Air ~ % ./7.1
Enter density: 7
Enter volume: 89
Mass = 623.00
dhrupandey@DHRUVs-MacBook-Air ~ %
```

Compilation and execution evidence for Question 1. The three runs produce Mass = 2.00, 6.00, and 623.00.

Code Evaluation

Actual outputs match the expected outputs for all three test cases. Result: **PASS**.

Question 2 - Pressure from Force and Area

1. Problem Understanding

Input: Force and Area

Output: Pressure

Formula: $\text{Pressure} = \text{Force} / \text{Area}$

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Force=7, Area=6	Pressure = 1.17	Pressure = 1.17	Pass
2	Force=8, Area=56	Pressure = 0.14	Pressure = 0.14	Pass
3	Force=9, Area=6	Pressure = 1.50	Pressure = 1.50	Pass

3. Algorithm

1. Start.
2. Read force and area.
3. Calculate $\text{pressure} = \text{force} / \text{area}$.
4. Display pressure.
5. Stop.

Algorithm Evaluation

The calculated pressures are 1.17, 0.14, and 1.50 (rounded to two decimal places). They match the expected values, so the algorithm passes all three test cases.

4. C Program

No separate source-code screenshot for Question 2 was provided. The report therefore does not invent or reproduce a text-code version.

5-6. Compilation, Execution and Code Evaluation

```
dhruvpandey@DHRUVs-MacBook-Air ~ % clang 7.1.c -o7.1
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.1.c
zsh: permission denied: ./7.1.c
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 1
Mass = 2.00
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 3
Mass = 6.00
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.1
Enter density: 7
Enter volume: 89
Mass = 623.00
dhruvpandey@DHRUVs-MacBook-Air ~ % nano 7.2.c
dhruvpandey@DHRUVs-MacBook-Air ~ % clang 7.2.c -o7.2
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.2
Enter force: 7
Enter area: 6
Pressure = 1.17
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.2
Enter force: 8
Enter area: 56
Pressure = 0.14
dhruvpandey@DHRUVs-MacBook-Air ~ % ./7.2
Enter force: 9
Enter area: 6
Pressure = 1.50
dhruvpandey@DHRUVs-MacBook-Air ~ %
```

Compilation and execution evidence for Question 2 (7.2.c -> 7.2).

Code Evaluation

The displayed results 1.17, 0.14, and 1.50 match the expected outputs. Result: **PASS**.

Question 3 - Cost of Electrical Energy

1. Problem Understanding

Input: Units Consumed and Cost per Unit

Output: Total Cost of Electrical Energy

Formula: Total Cost = Units Consumed x Cost per Unit

2. Test Case Table

Test Case	Input(s)	Expected Output(s)	Actual Output(s)	Result
1	Units=12, Cost/Unit=34	Total Cost = 408.00	Pressure = 0.35	Fail
2	Units=78, Cost/Unit=65	Total Cost = 5070.00	Pressure = 1.20	Fail
3	Units=88, Cost/Unit=67	Total Cost = 5896.00	Pressure = 1.31	Fail

3. Algorithm

1. Start.
2. Read units consumed and cost per unit.
3. Calculate total cost = units consumed x cost per unit.
4. Display total cost.
5. Stop.

Algorithm Evaluation

Using the required electrical-energy formula, the expected totals are 408.00, 5070.00, and 5896.00 for the three inputs above. The algorithm itself produces the required values.

4. C Program

No source-code screenshot showing the electrical-energy program was provided, so no plain-text code has been inserted.

5-6. Compilation, Execution and Code Evaluation

```
dhruvpandey@OHRUVs-MacBook-Air ~ % clang 7.1.c -o7.1
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1.c
zsh: permission denied: ./7.1.c
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 1
Mass = 2.00
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 3
Mass = 6.00
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1
Enter density: 7
Enter volume: 89
Mass = 623.00
dhruvpandey@OHRUVs-MacBook-Air ~ % nano 7.2.c
dhruvpandey@OHRUVs-MacBook-Air ~ % clang 7.2.c -o7.2
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.2
Enter force: 7
Enter area: 6
Pressure = 1.17
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.2
Enter force: 8
Enter area: 56
Pressure = 0.14
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.2
Enter force: 9
Enter area: 6
Pressure = 1.50
dhruvpandey@OHRUVs-MacBook-Air ~ % nano7.2.c
zsh: command not found: nano7.2.c
dhruvpandey@OHRUVs-MacBook-Air ~ % nano 7.2.c
dhruvpandey@OHRUVs-MacBook-Air ~ %
dhruvpandey@OHRUVs-MacBook-Air ~ % nano 7.3.c
dhruvpandey@OHRUVs-MacBook-Air ~ %
dhruvpandey@OHRUVs-MacBook-Air ~ % clang 7.2.c -o7.3
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.3
Enter force: 12
Enter area: 34
Pressure = 0.35
dhruvpandey@OHRUVs-MacBook-Air ~ %
```

Provided terminal screenshot while preparing Question 3.

```
dhruvpandey@OHRUVs-MacBook-Air ~ % clang 7.1.c -o7.1
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1.c
zsh: permission denied: ./7.1.c
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 1
Mass = 2.00
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1
Enter density: 2
Enter volume: 3
Mass = 6.00
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.1
Enter density: 7
Enter volume: 89
Mass = 623.00
dhruvpandey@OHRUVs-MacBook-Air ~ % nano 7.2.c
dhruvpandey@OHRUVs-MacBook-Air ~ % clang 7.2.c -o7.2
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.2
Enter force: 7
Enter area: 6
Pressure = 1.17
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.2
Enter force: 8
Enter area: 56
Pressure = 0.14
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.2
Enter force: 9
Enter area: 6
Pressure = 1.50
dhruvpandey@OHRUVs-MacBook-Air ~ % nano7.2.c
zsh: command not found: nano7.2.c
dhruvpandey@OHRUVs-MacBook-Air ~ % nano 7.2.c
dhruvpandey@OHRUVs-MacBook-Air ~ %
dhruvpandey@OHRUVs-MacBook-Air ~ % nano 7.3.c
dhruvpandey@OHRUVs-MacBook-Air ~ %
dhruvpandey@OHRUVs-MacBook-Air ~ % clang 7.2.c -o7.3
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.3
Enter force: 12
Enter area: 34
Pressure = 0.35
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.3
Enter force: 78
Enter area: 65
Pressure = 1.20
dhruvpandey@OHRUVs-MacBook-Air ~ % ./7.3
Enter force: 88
Enter area: 67
Pressure = 1.31
dhruvpandey@OHRUVs-MacBook-Air ~ %
```

Provided terminal screenshot containing three runs intended for Question 3.

Code Evaluation

Important: The supplied Question 3 screenshots show prompts for *force* and *area* and print *Pressure*. They also show the command **clang 7.2.c -o7.3**. Therefore, these screenshots demonstrate that the pressure source file was compiled as 7.3 rather than the required electrical-energy program. The actual outputs do not match the required Question 3 outputs, so the three code tests are marked **FAIL**. A correct Question 3 program screenshot and execution should replace these screenshots before final

submission.

Final Status

Question 1: PASS

Question 2: PASS

Question 3: Needs corrected program compilation/execution evidence.